

Mohamed Khaled

Subnets (3) Info							Last updated less than a minute ago	Actions	Create subnet
Find subnets by attribute or tag							< 1 >		
Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR				
DB-Subnet-C	subnet-0d1600619918bd52c	Available	vpc-030ce51729c41f3a2 VPC...	Off	192.168.1.0/24				
Public-Subnet-A	subnet-0d5511f71471c0ed8	Available	vpc-0b3272d99d98228a1 VPC...	Off	10.0.1.0/24				
App-Subnet-B	subnet-074f9a3e5fed15221	Available	vpc-04ddad6deb60c9f48 VPC...	Off	172.16.1.0/24				

Your VPCs (3) Info							Last updated less than a minute ago	Actions	Create VPC
Find VPCs by attribute or tag							< 1 >		
☐	Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR			
☐	VPC-C-DB	vpc-030ce51729c41f3a2	Available	Off	192.168.0.0/16	-			
☐	VPC-B-App	vpc-04ddad6deb60c9f48	Available	Off	172.16.0.0/16	-			
☐	VPC-A-Public	vpc-0b3272d99d98228a1	Available	Off	10.0.0.0/16	-			

Internet gateways (1) Info						Last updated less than a minute ago	Actions	Create internet gateway
Find internet gateways by attribute or tag						< 1 >		
☐	Name	Internet gateway ID	State	VPC ID	Owner			
☐	IGW-Lab17	igw-0be5f24294bef7790	Attached	vpc-0b3272d99d98228a1 VPC-A-Public	360832932700			

```
[ec2-user@ip-10-0-1-178 ~]$ ping -c 3 192.168.1.64
PING 192.168.1.64 (192.168.1.64) 56(84) bytes of data.
64 bytes from 192.168.1.64: icmp_seq=1 ttl=254 time=1.03 ms
64 bytes from 192.168.1.64: icmp_seq=2 ttl=254 time=0.622 ms
64 bytes from 192.168.1.64: icmp_seq=3 ttl=254 time=0.695 ms

--- 192.168.1.64 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2027ms
rtt min/avg/max/mdev = 0.622/0.785/1.038/0.181 ms
[ec2-user@ip-10-0-1-178 ~]$ ping -c 3 172.16.1.214
PING 172.16.1.214 (172.16.1.214) 56(84) bytes of data.
64 bytes from 172.16.1.214: icmp_seq=1 ttl=254 time=0.830 ms
64 bytes from 172.16.1.214: icmp_seq=2 ttl=254 time=0.653 ms
64 bytes from 172.16.1.214: icmp_seq=3 ttl=254 time=0.661 ms
```

```
[ec2-user@ip-10-0-1-178 ~]$ ping -c 3 192.168.1.64
PING 192.168.1.64 (192.168.1.64) 56(84) bytes of data.
64 bytes from 192.168.1.64: icmp_seq=1 ttl=254 time=1.03 ms
64 bytes from 192.168.1.64: icmp_seq=2 ttl=254 time=0.622 ms
64 bytes from 192.168.1.64: icmp_seq=3 ttl=254 time=0.695 ms

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[ec2-user@ip-10-0-1-178 ~]$ ping -c 3 172.16.1.214
PING 172.16.1.214 (172.16.1.214) 56(84) bytes of data.
64 bytes from 172.16.1.214: icmp_seq=1 ttl=254 time=0.830 ms
64 bytes from 172.16.1.214: icmp_seq=2 ttl=254 time=0.653 ms
64 bytes from 172.16.1.214: icmp_seq=3 ttl=254 time=0.661 ms

--- 172.16.1.214 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2026ms
rtt min/avg/max/mdev = 0.653/0.714/0.830/0.087 ms
[ec2-user@ip-10-0-1-178 ~]$
```

```
[ec2-user@ip-10-0-1-178 ~]$ ping -c 3 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=1.16 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=117 time=1.18 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=117 time=1.15 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 1.157/1.166/1.183/0.041 ms
[ec2-user@ip-10-0-1-178 ~]$ nslookup google.com
Server:          10.0.0.2
Address:         10.0.0.2#53

Non-authoritative answer:
Name:   google.com
Address: 64.233.180.113
Name:   google.com
Address: 64.233.180.138
Name:   google.com
Address: 64.233.180.139
Name:   google.com
Address: 64.233.180.100
Name:   google.com
Address: 64.233.180.101
Name:   google.com
```

Account ID: 3608-3293-2700

mohamed khaled

United States (N. Virginia)

[Alt+S]

Search

Amazon Linux 2

AL2 End of Life is 2026-06-30.

A newer version of Amazon Linux is available!

Amazon Linux 2023, GA and supported until 2028-03-15.

<https://aws.amazon.com/linux/amazon-linux-2023/>

[ec2-user@ip-10-0-1-178 ~]\$

CloudShell

us-east-1

Actions

Elastic IP addresses

Elastic IP addresses (1)

Find elastic IP addresses by attribute or tag

Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record
NAT-EIP-Lab17	3.208.250.167	Public IP	eipalloc-075a1af30c17c6057	-

Allocate Elastic IP address

Security Groups > sg-0e775e50bd7b4a2bf - bastion-sg

sg-0e775e50bd7b4a2bf - bastion-sg

Actions

Details

Security group name

bastion-sg

Security group ID

sg-0e775e50bd7b4a2bf

Description

Bastion SG

VPC ID

vpc-0b3272d99d98228a1

Owner

360832932700

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Outbound rules (1)

Manage tags

Edit outbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-0d6b76a1f0513a538	IPv4	All traffic	All	All

Security Groups > sg-0e775e50bd7b4a2bf - bastion-sg

sg-0e775e50bd7b4a2bf - bastion-sg

Details

Security group name

bastion-sg

Security group ID

sg-0e775e50bd7b4a2bf

Description

Bastion SG

VPC ID

vpc-0b3272d99d98228a1

Owner

360832932700

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (1)

Search

Name

Security group rule ID

IP version

Type

Protocol

Port range

-

sg-035b3c2ad96631218

IPv4

SSH

TCP

22

Security Groups (6)

Find security groups by attribute or tag

Name

Security group ID

Security group name

VPC ID

Description

-

sg-0201f687883bb6fb3

default

vpc-030ce51729c41f3a2

default VP

-

sg-074208c6ee918dbfd

app-sg

vpc-04ddad6deb60c9f48

App SG

-

sg-08984784964073649

default

vpc-04ddad6deb60c9f48

default VP

-

sg-065fec4c1c977db72

default

vpc-0b3272d99d98228a1

default VP

-

sg-0e775e50bd7b4a2bf

bastion-sg

vpc-0b3272d99d98228a1

Bastion SG

-

sg-086139e56c1a1d3d7

db-sg

vpc-030ce51729c41f3a2

DB SG

Instances (3)

Find Instance by attribute or tag (case-sensitive)

Name

Instance ID

Instance state

Instance type

Status check

Alarm status

Availability Zone

Bastion-Host-Lab17

i-09e1aee9824146050

Running

t2.micro

2/2 checks passec

View alarms +

us-east-1a

DB-Server-Lab17

i-0606a242f30b94033

Running

t2.micro

2/2 checks passec

View alarms +

us-east-1a

App-Server-Lab17

i-0b8b7a2e4056587cb

Running

t2.micro

2/2 checks passec

View alarms +

us-east-1a

NAT gateways (1)

Find NAT gateways by attribute or tag

Name

NAT gateway ID

Connectivity...

State

State message

Primary public IP...

Primary p

NAT-Lab17

nat-05c7ad999beb7aa486

Public

Available

-

3.208.250.167

10.0.1.226

Internet gateways (1)

Find internet gateways by attribute or tag

Name

Internet gateway ID

State

VPC ID

Owner

IGW-Lab17

igw-0be5f24294bef7790

Attached

vpc-0b3272d99d98228a1 | VPC-A-Public

360832932700

You can visualize and monitor your Transit Gateway(s) from the [AWS Network Manager](#). Register your Transit Gateway by creating a global network to get started.

Transit gateway attachments (3)

Info

Actions

Create transit gateway attachment

Find transit gateway attachment by attribute or tag

< 1 >

☐	Name	Transit gateway attachment ID	Transit gateway ID	State	Resource t...	Resource ID
☐	TGW-Attach-VPC-B	tgw-attach-01dfb7b6268f441d8	tgw-0f670212fa7895afa	Available	VPC	vpc-04ddad6debf
☐	TGW-Attach-VPC-A	tgw-attach-046fd59d430ae4c5e	tgw-0f670212fa7895afa	Available	VPC	vpc-0b3272d99d
☐	TGW-Attach-VPC-C	tgw-attach-062a8f04efa5d0064	tgw-0f670212fa7895afa	Available	VPC	vpc-030ce51729c

Select a transit gateway attachment

Transit gateways (1)

Info

Actions

Create transit gateway

Find transit gateway by attribute or tag

< 1 >

☐	Name	Transit gateway ID	Owner ID	State
☐	TGW-Lab17	tgw-0f670212fa7895afa	360832932700	Available

Route tables (1/3)

Info

Last updated 1 minute ago

Actions

Create route table

Find route tables by attribute or tag

< 1 >

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	-	rtb-0932e05b4000c34ea	-	-	Yes	vpc-030ce51729c41f3a
☐	-	rtb-0a1da5dbfa4418395	-	-	Yes	vpc-04ddad6deb60c9f
☒	-	rtb-0fe5d8e0160f34e5d	-	-	Yes	vpc-0b3272d99d98228

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Route tables > [rtb-0fe5d8e0160f34e5d](#)

Route table ID

[rtb-0fe5d8e0160f34e5d](#)

Main

☒ Yes

Explicit subnet associations

-

Edge associations

-

VPC

[vpc-0b3272d99d98228a1](#) | VPC-A-Public

Owner ID

360832932700

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (4)

Both

Edit routes

Filter routes

< 1 >

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0be5f24294bef7790	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table
172.16.0.0/16	tgw-0f670212fa7895afa	Active	No	Create Route
192.168.0.0/16	tgw-0f670212fa7895afa	Active	No	Create Route

rtb-0a1da5dbfa4418395

Route table ID

rtb-0a1da5dbfa4418395

VPC

vpc-04ddad6deb60c9f48 | VPC-B-App

Main

Yes

Owner ID

360832932700

Explicit subnet associations

-

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (4)

Both

Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	tgw-0f670212fa7895afa	Active	No	Create Route
10.0.0.0/16	tgw-0f670212fa7895afa	Active	No	Create Route
172.16.0.0/16	local	Active	No	Create Route Table
192.168.0.0/16	tgw-0f670212fa7895afa	Active	No	Create Route

0932e05b4000c34ea

Details

Info

Route table ID

rtb-0932e05b4000c34ea

VPC

vpc-030ce51729c41f3a2 | VPC-C-DB

Main

Yes

Owner ID

360832932700

Explicit subnet associations

-

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (3)

Both

Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	tgw-0f670212fa7895afa	Active	No	Create Route
172.16.0.0/16	tgw-0f670212fa7895afa	Active	No	Create Route
192.168.0.0/16	local	Active	No	Create Route Table

```
Nov 15 11:46:23 ip-10-0-1-178.ec2.internal systemd[1]: Starting The Apache HTTP Server...
Nov 15 11:46:23 ip-10-0-1-178.ec2.internal systemd[1]: Started The Apache HTTP Server.
[ec2-user@ip-10-0-1-178 ~]$
[ec2-user@ip-10-0-1-178 ~]$ echo "✅Apache installed and running on port 80"
✅Apache installed and running on port 80
[ec2-user@ip-10-0-1-178 ~]$ # من داخل الـ DB Server
[ec2-user@ip-10-0-1-178 ~]$
[ec2-user@ip-10-0-1-178 ~]$ # Install MySQL
[ec2-user@ip-10-0-1-178 ~]$ sudo yum update -y
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
No packages marked for update
[ec2-user@ip-10-0-1-178 ~]$ sudo yum install mysql-server -y
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
No package mysql-server available.
```

```
<p>VPC ID: 172.16.0.0/16</p>
[ec2-user@ip-10-0-1-178 ~]$ echo "<p>Transit Gateway Lab 17</p>" | sudo tee -a /var/www/html/index.html
<p>Transit Gateway Lab 17</p>
[ec2-user@ip-10-0-1-178 ~]$
[ec2-user@ip-10-0-1-178 ~]$ # Start Apache
[ec2-user@ip-10-0-1-178 ~]$ sudo systemctl start httpd
[ec2-user@ip-10-0-1-178 ~]$ sudo systemctl enable httpd
Created symlink from /etc/systemd/system/multi-user.target.wants/httpd.service to /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-10-0-1-178 ~]$
[ec2-user@ip-10-0-1-178 ~]$ # Check status
[ec2-user@ip-10-0-1-178 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; vendor preset: disabled)
   Active: active (running) since Sat 2025-11-15 11:46:23 UTC; 367ms ago
     Docs: man:httpd.service(8)
```

```
64 bytes from 172.16.1.214: icmp_seq=1 ttl=254 time=0.830 ms
64 bytes from 172.16.1.214: icmp_seq=2 ttl=254 time=0.653 ms
64 bytes from 172.16.1.214: icmp_seq=3 ttl=254 time=0.661 ms

--- 172.16.1.214 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2026ms
rtt min/avg/max/mdev = 0.653/0.714/0.830/0.087 ms
[ec2-user@ip-10-0-1-178 ~]$ ssh -o ConnectTimeout=5 -o StrictHostKeyChecking=no ec2-user@192.168.1.64
Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-10-0-1-178 ~]$ timeout 5 bash -c "</dev/tcp/192.168.1.64/3306" && echo "🟢 DB MySQL port 3306 is OPEN" || echo "🔴 DB MySQL port 3306 is CLOSED (normal if MySQL not installed)"
bash: connect: Connection refused
bash: /dev/tcp/192.168.1.64/3306: Connection refused
🔴 DB MySQL port 3306 is CLOSED (normal if MySQL not installed)
[ec2-user@ip-10-0-1-178 ~]$
[ec2-user@ip-10-0-1-178 ~]$ # من داخل الـ App Server (EC2 Instance Connect)
```